

AGENTS.md — novaecopia Operational Boundaries & Execution Rules

Target Scope & Agent Roles

This document specifies the operational rules, model tiering, and hardware boundaries for all autonomous agents executing within [novaexopia/](#).

1. Multi-Model Tiering & Responsibilities

Agent Tier	Primary Model	Hardware Allocation	Assigned Role in novaecopia
Triage / Gatekeeper	Qwen 2.5 / 3.5 0.8B	CPU Little Cores	Parses incoming voice/text payloads; determines whether an action requires tool execution, shell invocation, or direct conversational response.
Local Executor	Qwen 3.5 9B Q4_0 / Gemma 4 12B QAT	Hexagon v79 NPU (Node Alpha)	Executes complex reasoning, code inspection, AST queries, and MCP tool calls. Must obey NPU swapping protocols.
Frontier Scripter	Claude 3.7 Sonnet / Opus (ECC / OpenRouter)	Cloud API Fallback	Handles massive refactoring, deep multi-file compilation, and training batch synthesis.

2. Hardware & Memory Invariants (Anti-LMK Rules)

1. Strict Single-Model NPU Invariant:

- Android's Low Memory Killer (LMK) enforces strict memory pressure ceilings. Never attempt to load two local LLMs concurrently.
- Any model transition must issue a **SWAP** command to `/dev/socket/npu_manager.sock` and await confirmation before loading the next backend.

2. Thermal Throttling Ceilings:

- Battery/SoC temperature ceiling is set to **42°C** for background tasks and **45°C** for interactive inference. If exceeded, agents must switch to cloud fallback or pause.

3. Elevated Shell Loopback Protocol:

- Never attempt raw unprivileged shell execution. Route commands through **WirelessAdbBridgeService** on `127.0.0.1:5555` or the Ktor WebSocket bridge on `ws://127.0.0.1:8080/shell` to retain UID 2000 (shell) capabilities without rooting.
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3. Tool & Skill Execution Guidelines

- **MCP Isolation:** All filesystem, SQLite, and PostgreSQL database queries must pass through the registered MCP servers in `mcp_connectors/`.
- **Sensory Deferral:** Audio capture (`voice_stream_daemon.py`) and screen capture (`screen_vision_bounds.json`) operate under `aeyre/` with priority yielding during heavy model inference.
- **Naming Rule:** The canonical name of this repository and subsystem is **novaecopia** (*NovaeCopia™*). Never use deprecated or hallucinated names (e.g., "Omni Claw" is strictly prohibited).
- **Manifest Mutation Mandate:** Every new script, tool, or profile created in `novaexopia` must be registered in `manifest.jsonl` immediately upon creation.